TRƯỜNG ĐẠI HỌC FPT

CAPSTONE PROJECT REGISTER

Class: **Duration time: from 01/01/2026 To 30/04/2026**

(*) Profession: <Software Engineer> **Specialty**: <ES> ☐ <IS> ☒ <JS> ☐

(*) Kinds of person make registers: Lecturer ☒ Students

1. Register information for supervisor (if have)

No.	Fullname	Phone	**E-Mail**	**Title**
Supervisor	Đỗ Phúc Thịnh	0847088880	ThinhDP2@fe.edu.vn	Mr

2. Register information for students (if have)

	Full name	**Student code**	**Phone**	**E-mail**	**Role in Group**
	Phạm Đình Quốc Thịnh	SE171589	0842918005	thinhpdqse171589@fpt.edu.vn	Leader
	Nguyễn Nhật Nam	SE182539	0704656071	namnnse182539@fpt.edu.vn	Member
	Huỳnh Nguyễn Quế Trân	SE183031	0366856494	tranhnqse183031@fpt.edu.vn	Member
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3. Register content of Capstone Project

(*) 3.1. Capstone Project name:

- English: BoneVisQA - Interactive Visual Question Answering System for Bone Diseases to Support Medical Students in Diagnostic Practice
- Vietnamese: BoneVisQA - Hệ thống hỏi đáp trực quan về các bệnh lý xương hỗ trợ sinh viên y khoa thực hành chẩn đoán
- Abbreviation: BoneVisQA

a. Context:

In medical education, especially in orthopedics and musculoskeletal imaging, medical students need extensive exposure to and practice in interpreting X-ray, CT, MRI, and musculoskeletal ultrasound images. However:

- Access to a systematic set of representative image-based clinical cases is limited and heavily dependent on hospitals and individual clinical instructors.
- Students often learn in a fragmented, memorization-based manner (lesion names, diagnostic criteria) instead of linking knowledge to visual patterns and structured descriptions.
- There is currently no digital tool that allows students to ask questions directly on images (for example: "What condition does this lesion suggest?", "Which imaging sign is most important here?") and receive answers accompanied by references.
- Existing AI-based diagnostic support tools in the field of bone and joint diseases mainly target clinicians, with limited focus on educational objectives for students and with little transparency regarding knowledge sources or reasoning processes.

In this context, the project proposes to build a **multi-platform (web and mobile) interactive visual question answering system** (visual Q&A), in which students can view or upload X-ray/CT/MRI images of bones, ask questions about diagnosis and key imaging signs; the system combines AI-based image processing models and **Retrieval-Augmented Generation (RAG)** over a bone disease knowledge base to generate answers with cited sources, serving learning and diagnostic practice purposes.

b. Proposed Solutions

The BoneQA system will be developed with the following main components and characteristics:

Web platform for interactive Q&A on bone images

The system allows students to:

- Log in, manage personal profiles, and track learning progress.
- Select sample cases from a case library or upload bone images.
- Ask free-form questions about the images, for example:
  - "What type of fracture does this lesion suggest?"
  - "How to differentiate osteoarthritis from rheumatoid arthritis in this case?"
- Interact visually with images: zoom in/out and annotate suspicious regions on the image to be sent together with the question.

Mobile application to support learning anytime, anywhere

- Provide a mobile application for medical students.
- Allow login and synchronization of learning progress with the web system.
- Enable viewing of case lists and details on mobile devices, with touch operations for zooming and panning images.
- Allow students to ask visual questions directly on images during hospital rotations or clinical practice sessions.
- Support short quiz sessions (micro-learning) with bone images and corresponding questions.
- Provide notifications about new cases, new quizzes, or learning tasks assigned by lecturers.

Bone image processing and information extraction module

Use AI models to:

- Detect regions suspected of containing lesions on bone images (e.g., fractures, bone defects, osteoporosis, etc.).

- Extract image features to support:
    - Retrieval of similar cases in the case library.
    - Improving the relevance of document retrieval in the RAG mechanism.

Knowledge base on bone diseases and RAG mechanism

Build a knowledge base from:

- Open-access materials on bone diseases, textbooks, and basic guidelines (for educational purposes only, not for direct treatment decision-making).
- A set of sample cases (images + descriptions + diagnoses + key imaging signs that should be remembered).

Apply **Retrieval-Augmented Generation (RAG)**:

- Encode (embed) text segments and case descriptions into a vector database.
- When a student asks a question accompanied by an image, the system combines the question embedding and image features to retrieve the most relevant documents/cases.
- A generative language model (LLM) produces answers based on the retrieved context, with structured explanations and cited sources.

Role of Knowledge Base Manager (Content Curator / Knowledge Base Manager):

- Select, update, organize, and tag documents in the knowledge base.
- Trigger re-indexing when documents are added or modified, ensuring data quality for the RAG module.

Learning-oriented interface for medical students

Answers are presented in a structure that supports learning:

- Suggested main diagnosis.
- Important differential diagnoses.
- Key imaging signs to focus on.
- Reflective questions for student self-assessment.
- Links or citations to references from the knowledge base.

Support a **quiz mode**: the system proactively asks questions about a lesion region on an image; students answer, then the system displays the correct answer and explanation. This mode is optimized for both web and mobile interfaces.

Review workflow with lecturers/clinical experts

- Allow lecturers/clinical experts to review selected student questions and system-generated answers, assess appropriateness, edit content, and verify cases and explanations.
- Provide statistics on common mistakes in student answers to help lecturers adjust teaching methods and update training content.

c. Functional requirement

The system is expected to support the following main roles:

- Medical Student
- Clinical Expert / Reviewer
- Lecturer / Course Coordinator
- Content Curator / Knowledge Base Manager
- System Administrator

Web/Mobile portal for Students (Medical Student)

- Register, log in, and update personal information (school, cohort, major).

- View a list of sample bone cases categorized by:
    - Location (long bones, spine, knee, hip, shoulder, wrist, ankle, etc.).
    - Type of lesion (fracture, dislocation, degenerative disease, bone tumor, osteomyelitis, etc.).
    - Difficulty level (basic, intermediate, advanced).
- Open a case:
    - View images (X-ray/CT/MRI) with zoom/pan operations on web and mobile.
    - Annotate regions of interest on the image.
- Ask free-form questions about the case:
    - Enter questions in Vietnamese and optionally attach a link to the annotated region on the image.
    - Submit the question for processing by the system (Image + RAG).
- Receive answers:
    - View structured explanations (suggested diagnosis, key findings, differential diagnoses, recommended readings).
    - View citations to documents and similar cases from the knowledge base.
- Practice in quiz mode:
    - The system displays bone images and short questions (multiple-choice or short-answer).
    - Students respond and then receive feedback and explanations.
    - On mobile, this can be implemented as short learning sessions (micro-quiz).
- Track learning progress:
    - Number of cases viewed, questions asked, quizzes completed, and correctness rate.
    - Statistics by topic (e.g., long bone fractures, spine lesions, degenerative diseases).

Web portal for Clinical Experts (Expert / Reviewer)

- Log in with expert privileges.
- Manage the case library:
    - Add new sample cases (anonymous images + description + diagnosis + key learning points).
    - Tag cases by specialty, bone location, lesion type, and difficulty level.
- View student questions and system-generated answers:
    - Filter by case, time, student, keyword.
    - Assess clinical accuracy and educational appropriateness.
    - Edit/upgrade answers, add explanations or clinical notes.
    - Mark cases and explanations as “approved”.
- Define quiz sets and scoring criteria; propose topic-based quizzes for each class.

Web portal for Lecturers / Course Coordinators

- Log in with lecturer / course coordinator privileges.
- Manage classes and clinical rotations:
    - Create classes (e.g., “Orthopedics – Class A – Cohort 2023”).
    - Assign students to classes; import class lists from files.
- Assign learning content:
    - Assign case sets to classes or student groups.
    - Configure quiz sessions (topic, number of questions, open/close time).
- Monitor learning outcomes:

    - View statistics by class/group/individual: number of cases studied, quiz scores by topic, common interpretation errors.
    - Export reports to support assessment and teaching improvement.
- Send announcements to students:
    - Announce new cases, new quizzes, or required tasks to be completed before clinical rotations.

Web portal for Content Curator / Knowledge Base Manager

- Log in with knowledge base manager privileges.
- Manage documents in the knowledge base supporting RAG:
    - Upload and update open-access documents (guidelines, textbooks, teaching materials) related to bone diseases.
    - Organize documents by topics and subtopics; assign tags (e.g., "fracture classification", "osteoarthritis", "osteoporosis").
    - Manage document versions and mark outdated documents for update.
- Control the indexing pipeline for RAG:
    - Trigger embedding generation and re-indexing when new documents are added or existing documents are updated.
    - View indexing logs and handle errors.
    - Adjust chunking strategies to improve retrieval quality.
- Ensure content quality:
    - Monitor which documents/segments are frequently retrieved.
    - Detect low-quality or outdated content and flag it for review by experts/lecturers.

Web portal for System Administrator (System Admin)

- Log in with administrator privileges.
- Manage user accounts:
    - View users by role (Student, Expert, Lecturer, Content Curator, Admin).
    - Activate/deactivate accounts.
    - Assign/revoke roles for each user.
- Manage cases and image data:
    - Control adding/editing/deleting cases; hide cases that have not been approved.
    - Ensure that image data satisfies anonymization and privacy requirements.
- Configure the system:
    - Set confidence thresholds to decide when to display answers directly or suggest "consult additional documents/lecturers".
    - Configure technical parameters of AI modules.
- Monitor system logs and summary reports:
    - Track system activity, errors, and basic usage statistics (e.g., number of questions, number of case views, load).

Non-functional requirement:

- **Usability:** User-friendly interface for all roles; Vietnamese support (with the possibility to extend to English).
- **Security:** Data security, especially for medical images; clear access control; secure communication channels.

- **Performance:** Reasonable response time and stable operation for a medium-scale number of samples and users.
- **Scalability:** Ability to scale when the number of images, projects, and users grows.

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(*) 3.2. Main proposal content (including result and product)

a. **Theory and practice (document):**

 Apply system analysis and design methods (UML, requirements specification, database design).

 Conduct a literature review on:

- Common bone diseases and their typical imaging features (at a high level, for educational purposes).
- Image-based question answering systems (visual question answering, content-based image retrieval).
- Retrieval-Augmented Generation (RAG) techniques for knowledge-intensive question answering.

Server-side technologies

- Backend:
  - Node.js, C# .NET, Java Spring Boot, or equivalent to build REST APIs.
- AI module (Image + RAG):
  - Python (FastAPI/Flask) to implement the bone image processing module (feature extraction, lesion suggestion) and the RAG module (embedding, document retrieval, LLM calls).
  - Use a vector database (FAISS/Chroma or equivalent) to store embeddings of documents and cases.
- Database:
  - PostgreSQL/MySQL with an ORM (Prisma/TypeORM/Sequelize) or equivalent.

Client-side technologies

- Web:
  - React/Next.js, TypeScript, CSS/Tailwind to build interfaces for Students, Experts, Lecturers, Content Curators, and Admins.
- Mobile:
  - React Native (or Flutter) to build a mobile application for medical students, synchronized with the backend, supporting case viewing, interactive Q&A, and quizzes on mobile devices.

b. **Products:**

 Web application for students, clinical experts, lecturers, content curators, and system administrators:

- Supports viewing bone cases, interactive Q&A, quizzes, case library management, knowledge base management, and explanatory content management.

 Mobile application for medical students:

- Supports login, case viewing, asking visual questions on images, quick quizzes, and tracking learning progress on mobile devices.

 REST APIs for:

- Managing users, roles, cases, questions–answers, classes, knowledge base, and statistics.
- Communicating with AI image-processing and RAG modules.

 AI demo module:

- Performs basic processing of bone images (feature extraction, suggestion of suspicious regions).
- RAG module to retrieve and generate answers based on the bone disease knowledge base.

 Complete documentation set:

- System analysis and design documents (UML, ERD, requirements specification).
- Database design and implementation documents.
- Test plan and test report.
- Installation guide and user manuals for each role.

c. **Proposed Tasks:**

 Task package 1:
Requirements analysis, literature review, business process modeling, UML and database design.
 Task package 2:
Develop backend and REST APIs for managing users, roles, cases, Q&A, classes, and statistics.
 Task package 3:
Develop the web portal and mobile application for students, experts, lecturers, content curators, and admins, including the image viewer and Q&A/quiz interfaces.
 Task package 4:
Integrate the bone image processing AI module and the RAG module, connect them to the knowledge base, and build the indexing and retrieval pipeline.
 Task package 5:
Test the system with a demo case set, refine the interface and interaction workflow, and complete documentation (report and user guides).

4. Other comments (propose all relative things if have).

Supervisor (If have) *(Sign and full name)* Đỗ Phúc Thịnh	HCM, date 18/11/2025 **On behalf of Registers** *(Sign and full name)* Trần Đức Thịnh